Omar Khursheed

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CAREER SUMMARY

Software Engineering graduate at UNSW with experience building AI-powered systems, enterprise SaaS platforms, and cloud-native applications. Passionate about product development, building scalable systems and leveraging AI tools for efficient development.

PROFESSIONAL EXPERIENCE

Full Stack Engineer July 2025 - February 2026
Bahrix (Startup), Haymarket, Sydney

- Engineered features for an enterprise SaaS platform supporting project management and workforce operations.
- Integrated a RAG-based retrieval tool across 1,000+ reports, reducing manual document search by 20%.
- Improved retrieval method through neural re-ranking and ensemble methods, improving Top-5 accuracy by 25%.
- Created scalable PostgreSQL schemas and client-agnostic data models supporting multi-tenant platform growth.
- Devised event-driven integrations with third-party client systems in React and JavaScript automating workflow.
- Developed integration and component test suites achieving 85%+ coverage and reducing regression defects.

It Consultant Intern January 2023 - February 2023
Ernst & Young (EY)

- Formulated cloud migration/deployment proposals aligned to client requirements.
- Built and refined pitch decks for oil & gas clients, achieving 4.5/5 stakeholder satisfaction.
- Managed IAM access controls (RBAC/least privilege) to meet client security policies.

TECHNICAL PROJECTS

Fake Image Generator Model

- Constructed a Generative Adversarial Network in Python with Tensorflow to generate fake images.
- Built a Data Loader module in Python leveraging openCV, processing 10k+ image files.
- Enhanced results by introducing progressive growing training technique, improving FID score by 25%.

Cancerous Nodule Detector
Honours Thesis Project, UNSW

- Trained Large CNN models on 4.8k+ radiologist-annotated lung CXRs in Python via Pytorch, attaining mAP50 of 0.828.
- Resolved class imbalance through augmentations, expanding training sample by 30% and improving recall by 18%.
- Diagnosed small nodule detection bottleneck, driving cross-domain adoption of aerial tiny-object detection methods.
- Extended YOLOv5 model with Depth-Wise Reshaping and Multi-Scale Attention, increasing recall by 40%.

UNSW Timetable Manager Web App

- Collaborated with a team of students contributing to a timetable management application serving 1,000+ students.
- Designed and created Backend layer in Python using Flask for efficient data storage and processing.
- Optimized API endpoints performing data retrieval from MongoDB servers managing 100+ users.
- Integrated E2E testing and observability packages in React with Typescript, increasing test coverage by 40%.

Disease Analysis Web App

- Designed and deployed an AWS-powered micro–service architecture for real-time global outbreak tracking.
- Automated data collection deploying python script using AWS lambda, scraping 100+ data points weekly.
- Optimized and monitored cloud deployments using IaaC tools Terraform, enhancing deployment efficiency.

EDUCATION

Bachelor of Software Engineering (Honours)
University of New South Wales (UNSW)

- Dean's List, Australia Global Scholarship Award
- Computer Science Society Projects Subcommittee, Islamic Society UNSW Treasurer

CERTIFICATIONS

Certified Cloud Practitioner - AWS.
Machine Learning - Stanford University.

AWARDS AND HONORS

- Best Student Lead Project Award (UNSW): Developed a timetable manager web app used by campus students.
- Optiver Project Competition (UNSW): Project finalist for developing a data analysis web app.